THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 0	She returns to the starting point.
2	**(d)** acceleration	Change in velocity per time.
3	**(b)** velocity, acceleration and displacement	Third equation.
4	**(c)** greater than or equal to the magnitude of displacement	Path length is at least the straight-line gap.
5	**(d)** displacement	Product of v and t.
6	**(d)** 10 m/s squared	g about 9.8.
7	**(b)** vector quantity	It has magnitude and direction.
8	**(d)** at rest	Distance does not change.
9	**(b)** 20 m/s	v = gt.
10	**(c)** scalar quantity	Only magnitude.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Distance: total path length, scalar (1); Displacement: shortest straight-line gap with direction, vector (1).	1+1
12	Rate of change of velocity (1); m/s squared (1).	1+1
13	Direction of velocity changes continuously (1); Change in velocity means acceleration (1).	1+1
14	Speed: distance/time, scalar (1); Velocity: displacement/time, vector (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Distance = 3 + 4 = 7 m (1); Displacement = sqrt(3 squared + 4 squared) (1); = 5 m (1).	1+1+1
16	Graph is a horizontal line (1); Area = 6 x 10 (1); = 60 m (1).	1+1+1
17	Distance = 50 km in 1 h (1); Average speed 50 km/h (1); Speedometer shows instantaneous speed (1).	1+1+1
	OR: v = at = 12 m/s (1); s = (1/2)at squared = 0.5 x 3 x 16 (1); = 24 m (1).	1+1+1
18	a = 20/10 = 2 m/s squared (1); s = (10 + 30)/2 x 10 (1); = 200 m (1).	1+1+1
	OR: v = u - gt gives 0 = 20 - 10t, t = 2 s (1); h = u squared / 2g = 400/20 (1); = 20 m (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	v = u + at from slope (1); s = area of trapezium = ut + (1/2)at squared (1); v squared = u squared + 2as by eliminating t (1); Meaning of symbols (1); Conditions: uniform acceleration (1).	1+1+1+1+1
	OR: Velocity along the tangent (1); Direction changes: accelerated (1); v = 2 x 3.14 x 0.5/2 (1); = 1.57 m/s (1); Speed constant (1).	1+1+1+1+1
20	Stage 1: a = 5 m/s squared (1); Stage 2: a = 0 (1); Stage 3: a = -6 m/s squared (1); Distance: 90 + 300 + 75 (1); = 465 m (1).	1+1+1+1+1
	OR: 72 km/h = 20 m/s; 54 km/h = 15 m/s (1); Relative speed 5 m/s (1); Distance to cover = 100 + 150 + 150 (1); = 400 m (1); Time = 80 s (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** uniform velocity	Velocity constant.
2	**(c)** zero	Instantaneously at rest.
3	**(a)** m/s	Metre per second.
4	**(b)** uniform acceleration	v increases uniformly.
5	**(b)** the direction keeps changing	Direction changes.
6	**(b)** m/s squared	Rate of change of velocity.
7	**(c)** 25	90 x 5/18.
8	**(a)** along the tangent	Tangent to the circle.
9	**(b)** 7 m	3 + 4.
10	**(c)** 2 m/s squared	(20 - 10)/5.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Acceleration (1); Rise in velocity divided by time (1).	1+1
12	v = u + at (1); s = ut + (1/2)at squared and v squared = u squared + 2as (1).	1+1
13	No (1); Straight-line gap cannot exceed the path length (1).	1+1
14	Displacement (1); Area of a rectangle = v x t (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	t = 20/5 = 4 s (1); s = v squared / 2a = 400/10 (1); = 40 m (1).	1+1+1
16	a = (4 - 8)/2 = -2 m/s squared (1); Time to stop = 4/2 (1); = 2 s (1).	1+1+1
17	Scalar has magnitude only; vector has magnitude and direction (1); Scalars: distance, speed, time (1); Vectors: displacement, velocity, acceleration (1).	1+1+1
	OR: Average speed = 400/5 = 80 km/h (1); Net displacement = 0 (1); Average velocity = 0 (1).	1+1+1
18	Motion along a circle with constant speed (1); Examples: tip of a fan blade, Earth around the Sun (1); Velocity direction keeps changing (1).	1+1+1
	OR: Magnitude = sqrt(25 + 144) = 13 m (1); tan(theta) = 12/5 (1); About 67 degrees north of east (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Distance along the path (1); Displacement from the start (1); Speed = distance/time (1); Velocity = displacement/time (1); Acceleration due to changing direction (1).	1+1+1+1+1
	OR: Measure the distance (1); Time the student walking (1); Average speed = distance/time (1); Errors: reaction time (1); Repeat and average (1).	1+1+1+1+1
20	t up = 4 s (1); h = 1600/20 = 80 m (1); Total time 8 s (1); Velocity on return 40 m/s downward (1); Symmetry of motion (1).	1+1+1+1+1
	OR: h = 0.5 x 10 x 16 = 80 m (1); v = 40 m/s (1); Distance in 3 s = 45 m (1); Last second = 35 m (1); Using s = (1/2)gt squared (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Describing Motion Around Us** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 2 pi r / T	Circumference / time.
2	**(d)** 5 m	Pythagoras: sqrt(9 + 16).
3	**(b)** 50 km/h	100/2.
4	**(c)** instantaneous speed	Speed at that instant.
5	**(a)** equal displacements in equal intervals of time in the same direction	Constant magnitude and direction.
6	**(c)** v = u + at	First equation.
7	**(d)** retardation (deceleration)	Velocity decreases.
8	**(d)** speed	Distance divided by time.
9	**(b)** s = ut + (1/2)at squared	Second equation.
10	**(a)** changes with time relative to a reference point	Motion is relative.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Uniform: equal distances in equal times (1); Non-uniform: unequal, e.g. a car in traffic (1).	1+1
12	72 x 1000/3600 (1); = 20 m/s (1).	1+1
13	a = (20 - 10)/5 (1); = 2 m/s squared (1).	1+1
14	Total distance / total time (1); 5 m/s (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	h = 0.5 x 10 x 9 = 45 m (1); v = gt = 30 m/s (1); Starts from rest (1).	1+1+1
16	Total distance 50 m (1); Total time 5 s (1); Average speed 10 m/s (1).	1+1+1
17	Slope increases with time (1); Velocity is increasing (1); The body is accelerating (1).	1+1+1
	OR: A straight line sloping upward through the origin (1); Take two points and find rise/run (1); Slope = speed (1).	1+1+1
18	a = (v - u)/t (1); at = v - u (1); v = u + at (1).	1+1+1
	OR: Linear: a car on a straight road (1); Circular: a merry-go-round (1); Oscillatory: a pendulum (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rest: horizontal s-t line (1); Uniform velocity: straight s-t line, horizontal v-t line (1); Uniform acceleration: straight rising v-t line (1); Retardation: falling v-t line (1); Slopes and areas (1).	1+1+1+1+1
	OR: Motion is relative (1); A passenger is at rest relative to the train (1); Moves relative to the platform (1); Trees seem to move backward (1); Describing motion needs a reference (1).	1+1+1+1+1
20	u = 5 m/s (1); v = 5 + 5 = 10 m/s (1); s1 = 75 m (1); Retardation = 0.5 m/s squared; s2 = 100 m (1); Total = 175 m (1).	1+1+1+1+1
	OR: Time going = 2 h; returning = 3 h (1); Total distance = 240 km (1); Total time 5 h (1); Average speed 48 km/h (1); Average speed is total distance over total time (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.